

Homework 5 Solution

1. Let $A, B \in M_n(\mathbb{F})$. Suppose that A, B are similar. Prove or disprove each of the following statements:

(a) It holds that $A + B, A - B$ are similar.

Solution: False. We choose $A = B = I_2$. As the similarity relation is reflexive (which means that every matrix is always similar to itself), then A, B are similar. But $A + B = 2I_2$ and $A - B = 0_{2 \times 2}$ are not similar, as the zero matrix can only be similar to itself.

(b) Let $1 \leq i, j \leq n$. Then

$$\sum_{k=1}^n (-1)^{i+k} [A]_{ik} M_{ik}^{(A)} = \sum_{k=1}^n (-1)^{k+j} [B]_{kj} M_{kj}^{(B)}$$

Solution: True. As A, B are similar then $\det(A) = \det(B)$. Thus

$$\sum_{k=1}^n (-1)^{i+k} [A]_{ik} M_{ik}^{(A)} \stackrel{\text{i'th row expansion}}{=} \det(A) = \det(B) \stackrel{\text{j'th column expansion}}{=} \sum_{k=1}^n (-1)^{k+j} [B]_{kj} M_{kj}^{(B)}$$

(c) It holds that $\text{trace}(A^2) = \text{trace}(B^2)$.

Solution: True. As A, B are similar, then we have seen in class that A^2, B^2 are similar. We also know that the traces two similar matrices are equal, and therefore $\text{trace}(A^2) = \text{trace}(B^2)$.

(d) It holds that $\text{Null } A = \text{Null } B$.

Solution: False. Choose $A = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $B = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}$. We have seen in HW4Q2(d) that A, B are similar. We now compute

$$\text{Null } A = \left\{ \vec{x} = \begin{pmatrix} x \\ y \end{pmatrix} : A\vec{x} = \vec{0} \right\} = \left\{ \vec{x} = \begin{pmatrix} x \\ y \end{pmatrix} : \begin{pmatrix} y \\ 0 \end{pmatrix} = \vec{0} \right\} = \left\{ \begin{pmatrix} x \\ 0 \end{pmatrix} : x \in \mathbb{R} \right\} = \text{Span} \left\{ \begin{pmatrix} 1 \\ 0 \end{pmatrix} \right\}$$

and

$$\text{Null } B = \left\{ \vec{x} = \begin{pmatrix} x \\ y \end{pmatrix} : B\vec{x} = \vec{0} \right\} = \left\{ \vec{x} = \begin{pmatrix} x \\ y \end{pmatrix} : \begin{pmatrix} 0 \\ x \end{pmatrix} = \vec{0} \right\} = \left\{ \begin{pmatrix} 0 \\ y \end{pmatrix} : y \in \mathbb{R} \right\} = \text{Span} \left\{ \begin{pmatrix} 0 \\ 1 \end{pmatrix} \right\}$$

Thus, $\text{Null } A \neq \text{Null } B$.

(e) Suppose further that A, B are invertible. It holds that $\text{adj}(A), \text{adj}(B)$ are similar.

Hint. use HW1Q3(b).

Solution:

True. As A, B are similar, then there exists an invertible matrix $P \in M_n(\mathbb{F})$ such that

$$B = P^{-1}AP$$

Applying the adjoint to both sides and using HW1Q3(b), we get

$$\text{adj}(B) = \text{adj}(P^{-1}AP) = \text{adj}(P) \text{adj}(A) \text{adj}(P^{-1})$$

Choose $Q = \text{adj}(P^{-1})$. Then applying the formula in HW1Q3(b) again, we have

$$Q \cdot \text{adj}(P) = \text{adj}(P^{-1}) \text{adj}(P) = \text{adj}(PP^{-1}) = \text{adj}(I_n) = I_n$$

Thus, Q is invertible and $Q^{-1} = \text{adj}(P)$. Replacing back we get

$$\text{adj}(B) = \text{adj}(P) \text{adj}(A) \text{adj}(P^{-1}) = Q^{-1} \text{adj}(A) Q$$

Thus, $\text{adj}(A), \text{adj}(B)$ are similar.

2. Let $A = \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix}, B = \begin{pmatrix} a & b \\ 4b & a \end{pmatrix} \in M_2(\mathbb{C})$. For which values of $a, b \in \mathbb{C}$ are the matrices similar?

Solution:

Let $a, b \in \mathbb{C}$ and suppose first that A, B are similar. Then they have the same trace and determinant. Thus

$$3 = \text{trace } A = \text{trace } B = 2a \Rightarrow a = \frac{3}{2}.$$

Also,

$$2 = \det A = \det B = a^2 - 4b^2.$$

Plugging the value $a = \frac{3}{2}$ into the above equation, we obtain

$$\frac{0}{4} - 4b^2 = 2 \Rightarrow b = \frac{1}{4} \quad \text{or} \quad b = -\frac{1}{4}.$$

We now check each case separately.

Case 1: $a = \frac{3}{2}, b = \frac{1}{4}$. Choose $P = \begin{pmatrix} 2 & 5 \\ 2 & 1 \end{pmatrix}$. Note that P is invertible, as $\det P = -8 \neq 0$. Note also that

$$\begin{aligned} A \cdot P &= \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & 5 \\ 2 & 1 \end{pmatrix} = \begin{pmatrix} 8 & 8 \\ 4 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 5 \\ 2 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{3}{2} & \frac{1}{4} \\ 1 & \frac{3}{2} \end{pmatrix} = P \cdot B \end{aligned}$$

Thus $AP = PB$ which implies that $P^{-1}AP = B$, so A, B are similar.

Case 2: $a = \frac{3}{2}, b = -\frac{1}{4}$. Choose $P = \begin{pmatrix} 2 & 7 \\ -2 & 1 \end{pmatrix}$. Note that P is invertible, as $\det P = 16 \neq 0$. Note also that

$$\begin{aligned} A \cdot P &= \begin{pmatrix} 1 & 3 \\ 0 & 2 \end{pmatrix} \cdot \begin{pmatrix} 2 & 7 \\ -2 & 1 \end{pmatrix} = \begin{pmatrix} -4 & 10 \\ -4 & 2 \end{pmatrix} \\ &= \begin{pmatrix} 2 & 7 \\ -2 & 1 \end{pmatrix} \cdot \begin{pmatrix} \frac{3}{2} & -\frac{1}{4} \\ -1 & \frac{3}{2} \end{pmatrix} = P \cdot B \end{aligned}$$

Thus $AP = PB$ which implies that $P^{-1}AP = B$, so A, B are similar.

We conclude that A, B are similar if and only if $a = \frac{3}{2}$ and $b = \frac{1}{4}$ or $b = -\frac{1}{4}$.

3. Let $A, B \in M_n(\mathbb{F})$.

(a) Suppose that A is invertible. Prove that AB and BA are similar.

Solution: We choose $P = A$. Note that P is invertible because A is invertible. It holds that

$$P^{-1}(AB)P = A^{-1}(AB)A = (A^{-1}A)BA = I_n BA = BA$$

Thus, AB and BA are similar.

(b) Give an example where AB and BA are not similar.

Solution: We take

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \quad \text{and} \quad B = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$$

Note that

$$AB = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} = B$$

and

$$BA = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} 0 & 0 \\ 0 & 0 \end{pmatrix}$$

Therefore AB and BA are not similar (we have seen that the zero matrix is only similar to itself).

4. Let V be a vector space over $\mathbb{R}$ such that $\dim(V) = 2$. Let $\mathcal{A} = (v_1, v_2)$ be an ordered basis of V and let $T : V \rightarrow V$ be a linear transformation such that $[T]_{\mathcal{A}}^{\mathcal{A}} = \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$. Let $\mathcal{B} = (w_1, w_2)$ be an ordered basis of V for which

$$[T]_{\mathcal{A}}^{\mathcal{B}} = \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}$$

Prove that there is no ordered basis $\mathcal{D} = (u_1, u_2)$ for which the matrices $[T]_{\mathcal{D}}^{\mathcal{B}}$ and $[T]_{\mathcal{D}}^{\mathcal{A}}$ are similar.

Note: $[T]_{\mathcal{A}}^{\mathcal{B}}$ is the representation matrix of T from $\mathcal{B}$ to $\mathcal{A}$.

Solution:

ATC that there exists such a basis $\mathcal{D}$. Denote $P = [id_V]_{\mathcal{D}}^{\mathcal{A}}$. We know that P is invertible and therefore $\det P \neq 0$. Note that

$$[T]_{\mathcal{D}}^{\mathcal{A}} = [id_V \circ T]_{\mathcal{D}}^{\mathcal{A}} = [id_V]_{\mathcal{D}}^{\mathcal{A}} \cdot [T]_{\mathcal{A}}^{\mathcal{A}} = P \cdot \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix}$$

and

$$[T]_{\mathcal{D}}^{\mathcal{B}} = [id_V \circ T]_{\mathcal{D}}^{\mathcal{B}} = [id_V]_{\mathcal{D}}^{\mathcal{A}} \cdot [T]_{\mathcal{A}}^{\mathcal{B}} = P \cdot \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}.$$

As similar matrices have the same determinant (and we know $\det(AB) = \det A \cdot \det B$) then

$$\det P \cdot \det \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} = \det [T]_{\mathcal{D}}^{\mathcal{A}} = \det [T]_{\mathcal{D}}^{\mathcal{B}} = \det P \cdot \det \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix}$$

Dividing by $\det P \neq 0$, we obtain

$$-7 = \det \begin{pmatrix} 1 & 2 \\ 3 & -1 \end{pmatrix} = \det \begin{pmatrix} 3 & 2 \\ -1 & 4 \end{pmatrix} = 14.$$

Contradiction.